

XronoMorph: Investigating Paths Through Rhythmic Space

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Abstract XronoMorph is a musical loop generator that opens up two huge spaces of unusual and interesting polyphonic rhythms: *perfectly balanced rhythms* and *well-formed rhythms*. These are rhythms that would often be hard to create in alternative software applications or with traditional musical notation. In this chapter, I explain the algorithmic principles used to generate the loops and how these principles have been parameterized and visualized to facilitate the exploration of paths within these two rhythmic spaces.

Introduction

XronoMorph¹ is a software application designed to facilitate the creation of musically interesting looped rhythms and melodic hockets.² The loops are interesting for at least four reasons. First, they are frequently aesthetically pleasing. Secondly, they are often novel; in part, because they would be hard to create with conventional software or musical notation, or to perform manually. Thirdly, they are multileveled in that each rhythm comprises many interweaving musical voices. Lastly, each individual rhythmic level is founded on the same organizational principle as all levels combined: arguably, it is this “as above, so below” property that gives aesthetic unity to the rhythms; which takes us back to the first point.

XronoMorph uses two mathematical principles to organize the timings of events and the distribution of those events into multiple levels: *perfect balance*, which is a novel generalization of the *polyrhythms* common in, for example, sub-Saharan African music, and *well-formedness*, which generalizes the *additive rhythms* common in, for example, sub-Saharan and Balkan folk rhythms. Such rhythms are relatively uncommon in mainstream Western popular and classical traditions.

¹ The name *XronoMorph* is derived from the Ancient Greek nouns χρόνος (*khronos*, “time”) and μορφή (*morphē*, “form, shape”) whilst also noting the recent English (Greek-derived) verb *morph*, which means “smooth change”. We leave its pronunciation to the reader!

² A *hocket* is a melody whose successive pitches are sounded by different musicians – a practice used in medieval vocal music and in contemporary African and gamelan music.

Perfect balance and well-formedness each allow the production of both *rational rhythms* and *irrational rhythms* (a rational rhythm fits into a regular metrical grid, an irrational rhythm does not). In Western music, rational rhythms are the most common. However, jazz (and some funk and some rock/pop) often employs “swung beats”, where off-beats are systematically delayed by an irrational ratio of the regular grid; these systematic deviations from regularity may contribute to their perceived groove (Kilchenmann & Senn, 2015).

Interestingly, pairs of perfectly balanced rational rhythms can be smoothly connected by a continuum of perfectly balanced irrational rhythms; similarly, pairs of well-formed rational rhythms can be smoothly connected by a continuum of well-formed irrational rhythms. This allows XronoMorph to smoothly morph between pairs of rational rhythms by moving along these irrational paths. This is a highly distinctive feature, which opens up fascinating rhythmic possibilities. Furthermore, the irrational patterns that result from these manipulations are considerably more complex than the familiar swing ratio adjustments available in many standard music production tools.

To facilitate investigation of points in these rhythmic spaces and paths between them, XronoMorph provides novel and intuitive parameterizations and visualizations that support reflective and immediate (live) interaction.³

In subsequent sections, I will detail perfect balance and well-formedness, their relationships to existing music and to possible new musical endeavours, and the parametrizations and visualizations that make them accessible. But before proceeding to these more detailed expositions, I first consider related work in the domains of rhythm software and general organizational principles for rhythm.

In so doing, this chapter both draws together and extends recent publications related to perfect balance, well-formedness and XronoMorph (Milne, et al., 2015; Milne & Dean, 2016; Milne, et al., 2016; Milne, 2018; Milne, et al., 2018). For the first time, I combine straightforward explanations of the underlying music-theoretical concepts with substantive discussion of their creative potential (generating traditional rhythms – Western and non-Western – whilst, crucially, generalizing from these into unfamiliar rhythmic territory that would be otherwise hard to compose or to perform manually). There is also a novel emphasis on how these theories interface with practical design issues for the software, such as choosing appropriate visualizations, sonifications, and controls that facilitate live and interactive engagement (an example of the latter is Rhythmotron – the XronoMorph-based rhythm robot described in the final section). To further illustrate the concepts and demonstrate their creative possibilities in live performance, a number of new video examples are provided.

³ Additional organizational principles, parametrizations and visualizations will likely be added in future versions.

Related Software

Perhaps the most widely-used software applications for producing new rhythmic loops are *step sequencers*. A step sequencer comprises a regular grid of toggles. Each row of toggles indicates a specific sound, and each column indicates a specific time. When a toggle is on, it makes the corresponding sound at the appropriate time. There are typically no more than 16 time locations, which limits the range of rhythms that can be created. But, within that constraint, they are flexible and allow rhythms to be easily constructed.

What they do not so easily afford is higher-level manipulation of rhythmic patterns. For this reason, in a performance setting, they are not ideal for making improvised rhythmic transitions in a performance setting and, even in a compositional setting, they require a prior understanding of how good rhythms are constructed. Although step sequencers often allow for swing ratios⁴ to be adjusted, they do not allow for the construction of more complicated irrational rhythms.

Music-focused coding environments such as Max, SuperCollider, Gibber, ixi lang, and Tidal allow a user to create their own high-level rhythmic manipulations,⁵ but this requires expertise in coding and deep theoretical knowledge of rhythmic principles.

This has opened a space for rhythm applications that facilitate higher-level manipulations of rhythms than those offered by step sequencers, whilst using visualizations and parameters that facilitate interaction with the underlying organizational principles without in-depth knowledge of them. Perhaps the most widely used organizational principle in recent rhythm software applications are Euclidean (maximally even) rhythms (Clough & Douthett, 1991; Toussaint, 2013). This principle has inspired SequenceApp, Rhythm Necklace, Euclidean Sequencer, and others. However, Euclidean rhythms are defined only for rational rhythms hence all of the rhythms produced by these apps are restricted to the regular metrical grid; even more so than step sequencers with a swing ratio control.

In XronoMorph, we allow the user to break out of the grid, whilst still maintaining high-level control of rhythms. To do this, required developing and parameterizing organizational principles for rhythms that operate across continua of timing values. *Perfect balance* (introduced in (Milne, et al., 2015)) and *well-formedness* (first detailed in a rhythmic setting⁶ in (Milne & Dean, 2016)) both meet these criteria.

⁴ Changing the “swing ratio” usually delays every second or every fourth column so they no longer sound precisely halfway between their previous and following beats.

⁵ Indeed, XronoMorph is coded in Max and compiled as standalone applications for macOS and Windows.

⁶ Well-formedness was originally developed to explain musical scales (Carey & Clampitt, 1989).

XronoMorph

Prior to explaining the construction, parameterizations, and associated user-interface controls for the perfectly balanced and well-formed rhythms, it will be helpful to describe in more general terms how both types of rhythm are visualized and sonified, and the overall architecture of XronoMorph.

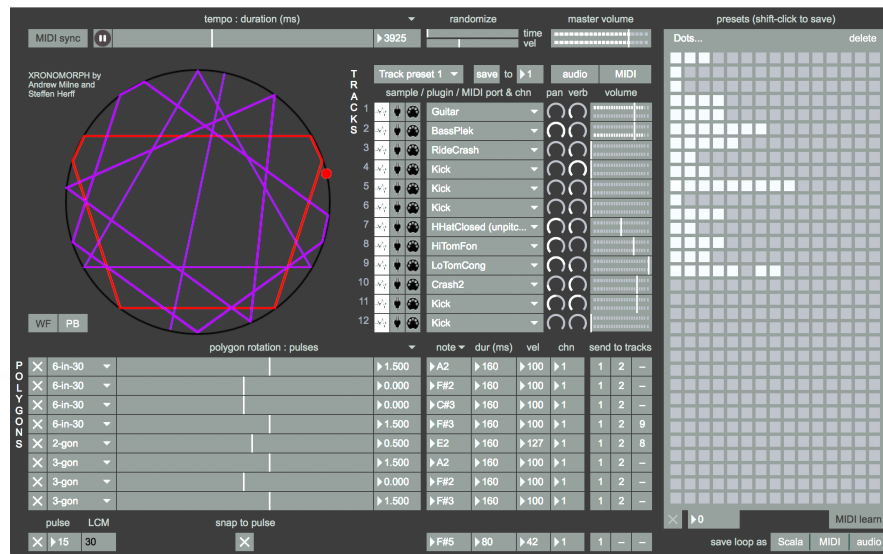


Fig 1. The interface of XronoMorph.

XronoMorph has two modes of operation: a perfectly balanced mode and a well-formed mode. These can be switched between using the associated “PB” and “WF” buttons, which change the parts of the user interface related to designing the rhythms. The remaining parts of the interface are unchanged and these are the focus of this subsection.

Both perfectly balanced and well-formed rhythms are, as previously mentioned, multilevel – they can comprise many rhythms that fully or partially interweave. For well-formed rhythms, there are up to six rhythmic levels; for perfectly balanced rhythms, there are up to nine rhythmic levels.

As shown in Figure 1, in the interface, each such level can be turned on or off: when it is on, every rhythmic event in that level is assigned a *single* pitch (with a MIDI pitch number or note name), a *single* velocity (0–127), and a *single* duration (in milliseconds). This constraint (just one set of note values per rhythmic level) may sound restrictive but it considerably simplifies the user interface and, in practice it seems to elicit creative results.

Each of these levels is visualized by a polygon inscribed in a circle. The circle is a natural and useful representation of periodicity, and it is commonly used to depict rhythms as in, for example, (London, 2004) and (Toussaint, 2013). A small

“playhead” disk rotates around the circle and, when it centres on the vertex of a polygon, it triggers that polygon’s pitch, velocity and duration values; the polygon also briefly lights up to show that it has been “triggered”. When the cursor hovers over any of the parameters of a level, the polygon also lights up to facilitate its identification. The use of inscribed polygons allows each different rhythmic level to be independently visualized.⁷

The pitches, durations, and velocities emanating from the rhythmic levels can be usefully thought of as a musical score. The next step is to “orchestrate” (or sonify) this score. Here, considerably more flexibility is allowed: an “orchestra” of twelve musical instruments (samples built into XronoMorph, or plugin software synthesizers, or external standalone software or hardware synthesizers) can be instantiated, panned, and changed in volume. Each instrument is numbered, and by entering up to three numbers in the routing field (labelled in Figure 1), each rhythmic level can be sent to up to three corresponding instruments. This allows for complex orchestrations to be easily created and adjusted during performance.

Another important aspect of the user interface is the large “presets” panel on the right-hand side. The entire state of XronoMorph can be saved to a box in this panel. This allows the user to easily switch between previously designed sequences that may involve changing rhythms, pitches, and orchestrations.

There are a few other global controls at the top of the interface. The large “tempo” slider sets the duration of the entire rhythmic *period* – the length of time in milliseconds – that the playhead completes a full circuit. There are sliders that control the degree of normally distributed random variation in the velocities and timings, which can result in more human-like performances (Kilchenmann & Senn, 2015). There is an overall volume control and controls related to MIDI synchronization with a standard digital audio workstation.

Perfectly Balanced Rhythms

The definition of *perfect balance* is simple: if each event in a rhythm is represented by a “weight” on a circle, that rhythm is perfectly balanced if its centre of gravity is at the centre of the circle.⁸ More concretely, we can think of the weight placed at a given time (position on the circle) as indicating the multiplicity of events occurring at that time: a weight of 0 would mean there were no events at that time, a weight of 1 would mean one event, a weight of 2 would mean two events (e.g., from two

⁷ Alternative ways of independently visualizing each different rhythmic level include using only small disks (no lines) on the circumference of the circle but colouring them according to the level, or using a set of concentric circles – one for each level. Every method has advantages and disadvantages: we chose inscribed polygons because they make the geometrical relationships clear (concentric rings distorts the relationships, colours are not obvious enough).

⁸ For a video demonstration of perfect balance using a physical bicycle wheel with attached weights, see <https://youtu.be/ipiogoqBibw>.

independent rhythmic streams), and so on. Clearly, in this context, a negative weight like -1 or -2 has no obvious interpretation.⁹

Despite the simplicity of this definition, perfectly balanced rhythms can be remarkably complex whilst also exhibiting a deep set of structural principles that come to light through advanced algebraic number (Galois) theory (Milne, et al., 2018). Here, I will explain these mathematical aspects, as much as possible, by analogy.

There are three useful mathematical theorems about perfectly balanced rhythms (Milne, et al., 2018), the first two of which are intuitively obvious. Together, they will allow us to understand how it is possible to create perfectly balanced rhythms that are remarkably complex yet highly organized. To highlight useful geometrical intuitions, I will typically refer to rhythms as *polygons*; for example, an equilateral triangle represents a rhythm with three equal interonset intervals, while an irregular triangle represents a rhythm with three unequal interonset intervals.

1. *All regular polygons (i.e., all isochronous rhythms) are perfectly balanced.* For instance, the centre of gravity of an equilateral triangle, a square, a regular pentagon, a regular hexagon, or so forth, is at its centre.
2. *The sum or difference of any two or more perfectly balanced polygons is also perfectly balanced.* All perfectly balanced polygons have the same centre-of-gravity (the centre of the circle), hence so will their sum. This means we can add together perfectly balanced rhythms to create new perfectly balanced rhythms: an “as above, so below” property, where each part has the same organizational property as the whole.
3. *All perfectly balanced rational rhythms can be constructed from the sum of regular polygons.* At first sight, this theorem may seem a little disappointing – it would seem to suggest that all perfectly balanced patterns are standard poly-rhythms such as those depicted in Figure 2.

⁹ Weights could also represent loudness, or probability of occurring, in which case real numbers are interpretable so long as they are nonnegative

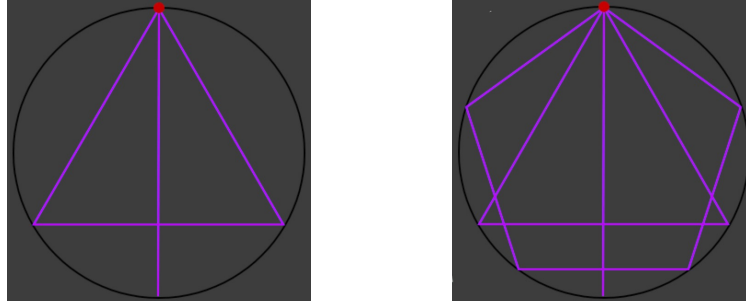


Fig. 2. Two “in-phase” polyrhythms. The left figure shows the familiar 2 against 3 polyrhythm; the right figure shows a less familiar but conceptually simple 2 against 3 against 5 polyrhythm. In these polyrhythms, there is one time location where all three regular rhythms sound simultaneously.

But, on closer inspection, the third theorem allows two important generalizations beyond the types of rhythm shown in Figure 2.

The first generalization derives from the simple observation that each of the summed regular polygons can be independently rotated so as to create more complex structures. This notion was perhaps first noted by Hofstadter (1985) – talking about Chopin’s frequent use of the three against two hemiola, he comments on how it seems like a natural generalization to displace the three-fold pulsation and the two-fold pulsation by putting them out of phase so that they never coincide – as shown in Figure 3. He notes that he has yet to find an example in the romantic repertoire of such rhythms. However, it seems these interweaving polyrhythms do occur in Sub-Saharan African music (Arom, 1991) (Milne, et al., 2018).

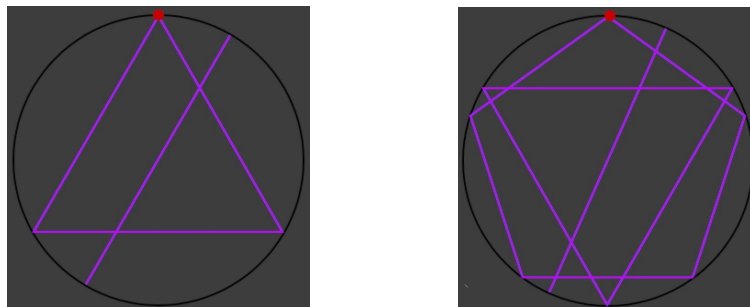


Fig. 3. These two polyrhythms are out-of-phase versions of those in Fig. 2. They comprise the same sets of individual rhythms (regular polygons), but they are rotated such that no two rhythms sound simultaneously. Arom (1991) calls such rhythms *strictly interweaving*.

The second generalization derives from noting that the regular polygons involved in the sum can have *negative weights*. Earlier, I stated that negative weights have no obvious musical interpretation but, so long as the final polygon that results from the sum has no negative weights, this ceases to be a problem. In such a case,

the negative weights play a role only in the construction of the rhythm but they are not manifest in its final form. Importantly, this method allows us to create perfectly balanced patterns that cannot be created simply by summing only positively-weighted regular polygons.

Clearly, to produce such shapes requires the negatively- and positively-weighted polygons to be precisely chosen and rotated prior to summation. Furthermore, these special rhythms can be formed only in metrical grids with a number of divisions that is equal to a multiple of at least three distinct primes, such as 30, 42, 60, 66, 70, 78, 84, 90, 102, 105, and so on. Figure 4 shows some intriguing examples of these rhythms, and their construction, in 30 and 42.

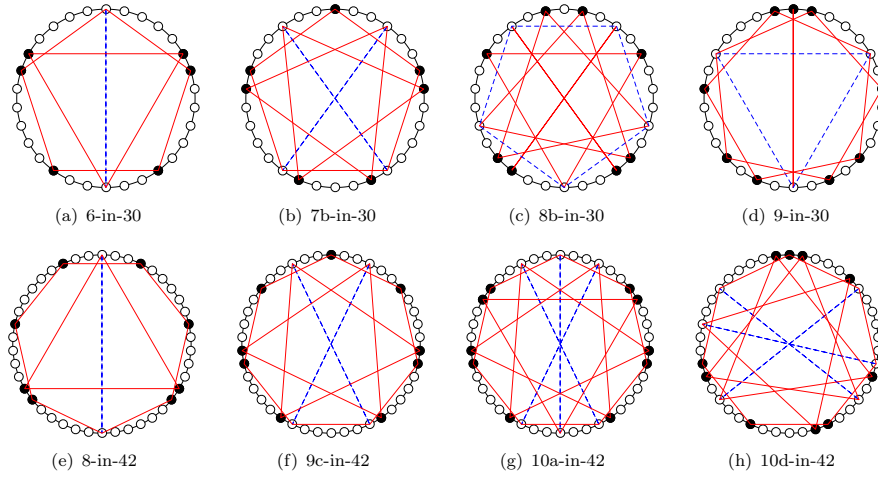


Fig. 4. Eight minimal primitives. Each pattern can be constructed by summing positively-weighted regular polygons (with solid lines) and negatively weighted regular polygons (with dashed lines). The minimality of these patterns means that none can be composed as a sum of only positively-weighted regular polygons. For example, the 6-in-30 pattern can be constructed from a “positive” regular pentagon, a “positive” regular triangle, and a “negative” *digon* (a two-vertex polygon which is simply a line) such that the latter “cancels out” one vertex from each of the former.

The fact that sums of perfectly balanced rhythms are also perfectly balanced (Theorem 1) suggests a parametrization for perfect balance: from all possible perfectly balanced rational rhythms we select, as “building blocks”, *minimal perfectly balanced rhythms*. A perfectly balanced rhythm is *minimal* if no subset of its events is also perfectly balanced. This means that all possible perfectly balanced rational rhythms can be created – by summation – from just these minimals (this is somewhat analogous to the way that all natural numbers can be generated from the prime numbers).

For example, a positively-weighted regular hexagon is not a minimal perfectly balanced rhythm because it can be “partitioned” into two perfectly balanced equilateral triangles at 180° . Conversely, an equilateral triangle is a minimal perfectly balanced pattern because it cannot be “partitioned” into a sum of smaller perfectly balanced patterns. Indeed, all regular polygons with a prime number of vertices are minimal perfectly balanced patterns, hence they are termed *regular minimals*.

But, what of the previously explained “special” perfectly balanced polygons that that must be built from at least one negative polygon (as in Fig. 4)? The task of finding which of these rhythms is minimal is computationally challenging – we have calculated all such *primitive minimals*¹⁰ in grids up to 102, but beyond this it has become computationally intractable (Milne, et al., 2018).¹¹

In XronoMorph, as shown in Figure 5, eight perfectly balanced rhythms can be simultaneously sounded, so the choice of these eight rhythms is the first set of user parameters. Each rhythm can be selected from a dropdown menu containing all regular minimals up to a 29-gon, and all 30 primitive minimals available in grids up to 42. A number of additional (non-prime) regular polygons are also included because these are frequently useful.

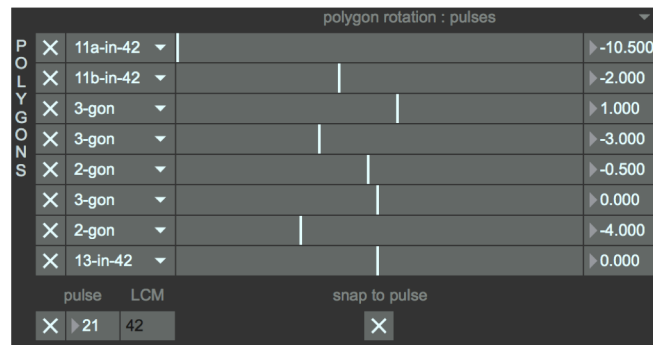


Fig. 5 Controls for each of the perfectly balanced polygons in a set. The sliders control their rotation (displayed on the right in pulses, degrees, radians, or turns). The menus on the left determine which perfectly balanced polygon is used. “Pulse” sets the number of the pulses in each period against which the other levels snap when rotated (when “snap to pulse” is enabled). “LCM” displays the smallest pulse value into which all of the selected polygons can fit. The pulse level can also be sounded.

The second set of user parameters is the individual rotation of each rhythm (recall that perfect balance is unaffected by rotation). Each rhythmic polygon can be freely rotated over a full turn (360°) but, by engaging the “snap-to-pulse” toggle (see Figure 5), rotations are forced to snap to a regular division of the circle. This regular division is set by the user in the adjacent “pulse” value. If this “pulse” value is

¹⁰ The term “primitive” refers to their lack of rotational symmetry.

¹¹ The computational methods we have used are detailed in (Milne, et al., 2018).

equivalent to, or some multiple of, the displayed “LCM” value,¹² every event will fall on one of these pulses. These regular pulses can be independently sonified as a ninth rhythmic level, often serving the musical role of binding the various polygons into a unified whole.

“Snap-to-pulse” ensures all resulting rhythms are, in totality, rational – much in the same way that “snap-to-grid” in a conventional MIDI sequencer’s piano roll ensures events are in a rational meter. Disengaging “snap-to-pulse” allows irrational perfectly balanced rhythms to be constructed.

In this way, polygons can be easily rotated during performance to produce interesting changes in rhythmic structure. Perfectly balanced patterns also seem to be very useful for creating melodic hockets: by assigning different pitches to different polygons, they frequently sound out half-heard or implicit melodic phrases or bass lines. An example of such a bassline can be heard in the perfectly balanced sections of *Babylon 19130* (<https://youtu.be/7EYjO9LYjYA>, 7:26–8:05 and 8:40–11:09): although each polygon produces only a single bass pitch, their interweaving creates an intricate melodic riff. With hockets, independently rotating the polygons changes the perceived riff: a sequence of hocketed bass riffs produced by different rotations of the same three polygons can be heard in the piece *Spider’s Eye* (<https://youtu.be/XPIITxfillC>).

Well-Formed Rhythms

Well-formed rhythms are interesting because they comprise a natural hierarchy of possibly interweaving rhythmic levels that are well-formed both individually and in sum. Well-formedness also has a relatively simple mathematical parameterization that can be directly controlled by a composer or performer. Before explaining how this rich hierarchy and its parametrization is created, it is first helpful to obtain a grounding in the notion of well-formedness by considering just a single rhythmic level.

Generating and Parameterizing a Single Well-Formed Rhythmic Level

A *single well-formed rhythm* can be simply defined: it is a rhythm with no more than two interonset intervals and these two interonset intervals are distributed as evenly as possible.

A way of visualizing this definition is through the use of the *cutting sequence* through a grid of squares (Figure 6). A sloping *cutting line* is drawn starting from a vertex of one of the squares. Going from left to right, the slope of the line determines if and when the sloping line will again pass through a grid vertex – if the slope is

¹² The LCM value is calculated from the user-selected polygons; it shows the smallest possible grid size that can contain all polygons’ vertices when they are snapped to that grid.

m/n , the line will pass through a vertex every m squares right/left and n squares up/down. Between these two vertex crossings, the line cuts through vertical and horizontal lines in the grid. If each of the $m - 1$ vertical crossings by v (for vertical), each of the $n - 1$ horizontal crossings is denoted by h (for horizontal), and the start and end of the line (the two vertex crossings) are denoted v and h , respectively, the resulting sequence of $m + n$ letters is known as a *Christoffel word* (Berstel, et al., 2008).

Every different rational slope of the cutting line – as defined by the two numbers m and n – produces a unique Christoffel word and, within each such Christoffel word, the two letters (v and h) are distributed as evenly as possible. This sequence of letters can be used to represent a sequence of any binary musical feature.

In XronoMorph, v represents a “long” interonset interval, and h represents a “short” interonset interval. The numbers m and n are two of the parameters that can be freely manipulated: they determine the number of long and short interonset intervals in the rhythm, while their ordering is automatically determined by the resulting Christoffel word. The parameters m and n , therefore, affect the gross structure of the rhythm – its “temporal contour”.

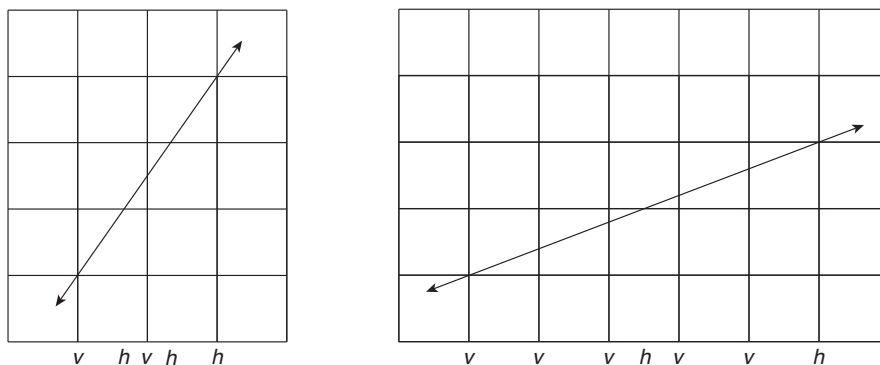


Fig. 6. Two cutting sequences. On the left is the cutting sequence for a line with slope $3/2$; the sequence of vertical (v) and horizontal (h) cuts spell out the Christoffel word $vhvhh$. Note that if v represents a “large” step and h a “small” step, this is the sequence of steps sizes in the familiar pentatonic scale. On the right is the cutting sequence for a line with slope $2/5$, which spell out the Christoffel word $vvvhvvh$, which corresponds to the sequence of step sizes in the diatonic scale (the Lydian mode). In XronoMorph, h and v refer to long and short interonset intervals in well-formed rhythms.

Of course, to actually sound a rhythm, it is necessary to know not just the sequence of long and short interonset intervals, but also their precise durations. This requires a novel parameterization, which is now detailed. If the duration of the “long” interonset interval is denoted L and the duration of the “short” interonset interval is denoted s , the notion of “large” and “small” is formalized by the constraint $s \leq L$. An additional constraint is that the sum of the lengths of all the interonset intervals equal a constant duration d , which is the duration of the rhythmic period as set by the user using the “tempo” slider described earlier. This means that

$mL + ns = d$ (recall that there are m long steps and n short steps). This second constraint means that, for any previously chosen numbers of long and short steps, the durations L and s must contravary; that is, as long interonset intervals get longer, the short interonset intervals must get shorter, and vice versa.

There are at least three ways to parameterize the durations L and s . First, the user could directly control L , with s being accordingly calculated; secondly, the user could directly control s , with L being accordingly calculated; thirdly, the user could control the ratio $r = L/s$ with the parameter t such that $r = 1/(1 - t)$. This means that when $t = 0$, $r = 1$ (which implies $L = s$), and when $t = 1$, $r \rightarrow \infty$ (which implies s has shrunk to zero size).¹³

In XronoMorph, we use the latter parameterization because, by symmetry, the distribution of rational r values across t is patterned and centred around $t = 0.5$, as shown in Figure 8 (top). The reason that rational values of r are of interest is because, when r is rational, so is the rhythm: in other words, when r is a ratio of any two whole numbers, the rhythmic events will always align with a regular metrical grid.¹⁴ Indeed, these rational well-formed rhythms include all possible Euclidean rhythms; they expand the palette by including a variety of non-Euclidean rhythms.

The value of r is also important because it affects not just the rhythmic level to which it is directly applied; it also affects the entire hierarchy of rhythms derived from that rhythm, as now described.

Generating and Parameterizing a Hierarchy of Well-Formed Rhythms

Given a well-formed rhythm with $r > 1$ (i.e., $L > s$), it is always possible to generate a *higher-level* well-formed rhythm that is denser (faster); this is done by inserting a new rhythmic event that consistently splits each long interonset interval into a new long and a new short interonset interval. Evidently, this creates a new rhythm with m more events than before.

However, to ensure the higher-level rhythm is also well-formed, each split must be made at a temporal location that ensures one of the new interonset intervals has the same duration as the previous rhythm's small interonset intervals (which have not been split). If this were not the case, the higher rhythmic level would have three different beat sizes and hence would not be well-formed, by definition.

This process is illustrated in Figure 7, which shows how a well-formed rhythmic level with 2 long beats and 1 short beat, is split into a rhythm with 2 long beats and 3 short beats, which is split into a rhythm with 5 long beats and 2 short beats, which

¹³ There are still other parametrizations that naturally derive from alternative ways of generating well-formed patterns through the use of the repeated iteration of an interval, where the size of the commonest interval (the generating interval) in the scale or rhythm is adjusted. For pitch-based scales this may be important because we may want this commonest interval to be itself consonant or a simple fraction of a consonant interval. This is the parameterization utilized in much microtonal music theory and in the Dynamic Tonality software synthesizers (<http://www.dynamictonality.com>).

¹⁴ This regular metrical grid divides the period into $mL + ns$ equal parts.

is split into a rhythm with 12 equally sized beats.¹⁵ Once a level – like this last one – is perfectly even (isochronous) it cannot be split any further and the hierarchy terminates.

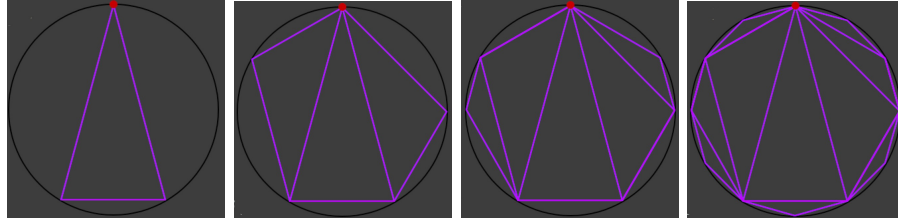


Fig. 7. A hierarchy of well-formed patterns. The lowest level rhythm, on the left, comprises two long interonset intervals and one short interonset interval. The next higher level splits the previous long intervals into a new long and short (one of whose durations matches the pre-existing short interval) to create a rhythm with 2 long and 3 short intervals. The next higher level splits the previous level's intervals into new long and short intervals to make a 5 long, 2 short rhythm. The fourth level is the last because it has reached an isochronous twelve step rhythm that is isochronous and, consequently, can no longer be further split.

In XronoMorph, there is a large slider – the *r*-slider – that is used to control the *r*-value of the *lowest* rhythmic level – see Figure 8. The *r*-values for all higher levels are automatically calculated and cannot be directly manipulated, but they are displayed in the interface – see Figure 9.¹⁶



Fig. 8. The *r*-slider – at the bottom of both figures – controls the long/short ratio of the lowest level rhythm. In the top figure, the numerical values show the *r*-slider positions that will produce an isochronous rhythm at some higher level; in this example, there is an isochronous rhythm with 18 events at the fourth level. In the bottom figure, the symbols show so called metallic ratios (see text for more details) that produce deeply nonisochronous rhythms. Any of the numbers or symbols above the slider can be clicked to move the slider that precise value. The *r*-slider value is also displayed, and can be entered, numerically on the right of the slider. Above that are toggles to show or hide the isochrony numbers or the various metallic ratios.

¹⁵ Note that this is the same pattern as the pentatonic, diatonic, and chromatic scales. Other well-formed patterns are quite different to this hierarchy.

¹⁶ These higher-level *r*-values are calculated as a function of the *r*-slider's value and *m* and *n*; the equations for this task are beyond the scope of this chapter but are detailed in (Milne & Dean, 2016).

		L	s	L/R	U/C	polygon : r-value (1–inf)	
POLYGONS	X	23	10	R	C		INF
	X	10	13	R	C		1.5
	X	10	3	R	U		2.
	X	3	7	R	C		1.286
	X	3	4	L	U		2.5
	X	3	1				3.5
	X	3	1				

Fig. 9. Controls for each of the polygons in a well-formed hierarchy. The numbers of large and small interonset intervals in the lowest level rhythm can be entered here (the ratio of their sizes is set by the r -slider in Fig. 8). The r -values for the higher levels are displayed on the right – these cannot be directly changed; they are automatically calculated from the r -slider. The U/C switches moves each rhythm between universal and complementary modes (see text for details); the L/R switches move each rhythm between left and right modes (also explained in the text).

As already implied, a higher-level well-formed rhythm can only be isochronous when the r -slider is set to a rational value (i.e., it is a ratio of whole numbers). The number of levels that must be ascended (from this lowest rhythmic level) to achieve isochrony depends upon where this r -value appears in the Stern-Brocot tree (Milne, et al., 2011).¹⁷ Note how in Figure 8, different numbers appear on different horizontal stripes. The value of each such number shows the number of events in the isochronous rhythm, its horizontal position shows the r -value required for the lowest rhythmic level, its vertical position shows on which higher level isochrony will occur. For example, if a number is found on the third line, the third level of the rhythmic hierarchy will be isochronous. To facilitate snapping the r -value to these precise rational values, any one of the numbers above the r -slider can be clicked on and the r -slider will snap to the appropriate position. This makes it possible to instantaneously switch between related but different rational rhythms: they are related because the lowest level always has the same sequence of long and short durations; they differ because some or all of the higher levels change and, crucially, the regular pulse that intersects with all rhythmic levels changes speed.

But rational r are not the only values of interest. If we randomly choose an irrational r -value, no rhythm will ever reach perfect isochrony. But, some higher levels may well come close to isochrony; close enough to be perceptually indistinguishable. This is because, for almost all irrational r , successively higher levels in the hierarchy have r -values that “wander” across the entire space of r -values and often come close to $r = 1$, which is isochrony.

There are, however, certain “special” values of r that ensure no higher-level rhythm ever approaches isochrony. This is because, at these special values, the r -values of all higher levels no longer wander; instead, they become “locked” to a

¹⁷ The Stern-Brocot tree is a systematic enumeration of the rational numbers independently discovered in the 19th century by the mathematician Moritz Stern and the watchmaker Achille Brocot. Stern’s focus was mathematical whereas Brocot’s focus was the specification of gear ratios for clock design. The tree provides a method to iteratively generate all rational numbers, in reduced form, exactly once.

single value, or to a repeated sequence of two values, or a repeated sequence of three values. These special r -values correspond to the so-called metallic ratios: the *golden ratio*, which is $\phi = (1 + \sqrt{5})/2 \approx 1.618$; the *silver ratio*, which is $\delta = (2 + \sqrt{8})/2 \approx 2.414$; the *bronze ratio*, which is $\sigma = (3 + \sqrt{13})/2 \approx 3.303$. This sequence of metallic ratios continues indefinitely. The golden section ensures the r -values of all higher levels are also in the golden section; the silver ratio ensures successively higher levels alternate between δ and $\delta - 1$; the bronze ratio ensures all successively higher levels sequence through the repeating cycle $\sigma, \sigma - 1, \sigma - 2$. In all cases, there is no r -value value that comes perceptibly close to 1 (which would be perfect isochrony); they are all heard as distinctly nonisochronous. We call such rhythms *deeply non-isochronous* rhythms.

In this way, the r -values can be used as a performative and compositional tool. The r -slider can be used to smoothly glide between isochronous rhythms, the fractions above the r -slider can be used to switch between rational rhythms, such that the overall rhythmic pattern is related (the base level always comprises the same number of large and small interonset intervals), but the upper levels may shift in structure, and the fastest isochronous level changes in speed, whilst the overall period of the rhythm remains constant. Switching directly to the irrational rhythms with the ϕ , δ , and σ symbols allows for deeply non-isochronous rhythms to be instantly accessed. These deeply non-isochronous rhythms typically have a “restless” or “tense” feel that can provide a contrast to the more “stable” isochronous rhythms; indeed, the transitions between the two types of rhythm can play a metrical role that is analogous to the use of tonality (e.g. consonance and dissonance and tonal function) to imbue tension and release.

A live musical performance that exemplifies some of these manipulations of the r -slider can be seen and heard in the middle section of “Primitive Minimals” (https://youtu.be/vinte4_ZL0o, 2:55–5:10): the clickable numbers and Greek letters above the r -slider (see Fig. 8) allow a performer to easily improvise jumps between related rational and deeply non-isochronous rhythms. Furthermore, as demonstrated in <https://youtu.be/-UAECGZhbFU> (1:10–1:50), the r -slider itself allows for smooth morphs between related rhythms to be easily performed.

As shown in Figure 9, there are two other parameters relevant to well-formed rhythms. Both of these parameters are applicable to each rhythmic level. The first of these switches each level between *universal* and *complementary* mode (shown as “U/C” in the interface). In the well-formed hierarchy pictured in Figure 10(a), note how each successive level has a rhythmic event at the same time as all previous levels. Clearly this results in considerable duplication of the lower levels’ events. By putting any level into complementary mode, all such duplications are removed so as to leave only complementary events. Due to a remarkable mathematical theorem (Amiot, 2007), these complementary levels are themselves also well-formed. As illustrated in Figure 10(b), by setting all rhythmic levels to complementary mode, it is possible to create a rhythmic structure comprising strictly interweaving (never coinciding) rhythmic events, where every level is well-formed and the rhythm made by the totality of such levels is also well-formed. This “as above, so

below” property is, as previously mentioned, also found in perfectly balanced rhythms.

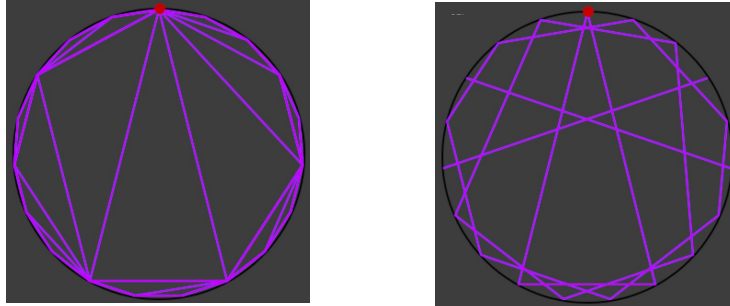


Fig. 10. On the left is a well-formed hierarchy with all levels in “universal” mode, which means that every level sounds at the same time as any lower level. On the right, all rhythms in the same well-formed hierarchy are set to “complementary” mode, which means that no two levels ever sound simultaneously – but all levels are still well-formed.

The remaining parameter for each rhythmic level – labelled L/R in the user interface – simply reverses the splitting order from long-short to short-long. This results in a more subtle variation in the rhythm that is useful for providing variety during performance.

A video demonstrating a number of the above mentioned features, such as successively building a well-formed hierarchy, thinning out a rhythm with the U/C controls, morphs along irrational paths between isochronous rhythms, and switching between rational rhythms can be viewed at <https://youtu.be/-UAECGZhbFU>.

Conclusion

I made the first prototypes of XronoMorph in 2014 – originally called MeanTimes (Milne & Dean, 2016) – to test whether the abstract concept of well-formedness and the newly developed, and similarly abstract, concept of perfect balance (Milne, et al., 2015) were viable principles for generating rhythms. Upon actually hearing these two types rhythms for the first time, it seemed to me quite clear that they opened doors to two remarkably interesting and aesthetically pleasing rhythmic spaces. Given the enthusiastic responses it has received from music producers and music enthusiasts (e.g., <http://www.dynamictonality.com/XMreviews.htm>), it seems I have not been alone in making this assessment.

The apparent success of XronoMorph is due not just to the mathematical principles that underlie its rhythms but also, as discussed in this chapter, to careful consideration of how to parameterize these rhythms, to visualize them, and to sonify them. These aspects were designed with music producers in mind – users who already have experience with music software and are familiar with basic notions of

signal flow and MIDI. But there is also scope for designing simplified versions more suitable for those with less musical experience or for children. Indeed, beyond its creative applications, we are currently using – in two pilot projects at schools in Sydney, Australia – simplified versions of XronoMorph to teach music with mathematics and mathematics with music.

Another interesting strategy, for artistic purposes, is to deliberately obscure the meaningful parametrizations: Rhythmotron is a robotic drum machine housed inside a repurposed upright piano (the original design sketch is shown in Figure 10). The robotic drumsticks are driven by XronoMorph, but the software (and its interface) are hidden. Instead, users interact with large hardware controls (four toggles, four momentary switches, six rotary knobs). Many of these controls are enigmatically labelled and are often mapped to combinations of XronoMorph's parameters. This means that adjusting any one of the hardware controls will change the rhythm – sometimes subtly, sometimes drastically – but the actual effect produced by a given manipulation will often be hard to anticipate.

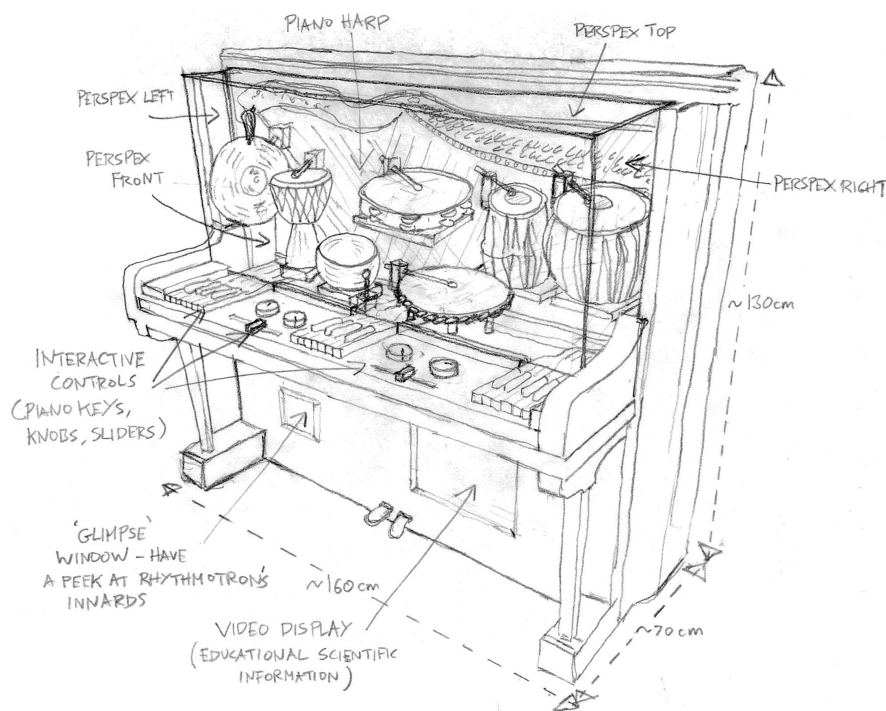


Fig. 10. The preliminary design sketch for Rhythmotron – a robotic drum machine housed inside an upright piano. The robots are driven by XronoMorph, which is controlled by hardware buttons and knobs. Rhythmotron was installed in Bungarribee Park, Western Sydney, at the ten-day CoLABS: Art and Science Collide event in October 2017.

Unsurprisingly, Rhythmotron was particularly popular with children who were enthused to hear (and see) how their knob-twists and turns created exciting and visceral beats – their parental overseers for the most part blissfully unaware that the resulting time signatures and irrational beats would challenge even the most experienced of musical performers!



Fig. 11. The completed Rhythmotron, freshly installed at CoLABS 2017 (see here for a video <https://youtu.be/IbJxN45naN8>).

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